Multimodal Difference Augmentation Learning for Remote Sensing Change Detection

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***Abstract*—Remote sensing change detection (RSCD) plays a crucial role in applications such as environmental monitoring and urban planning. With the emergence of foundational vision-language models like CLIP, there is growing interest in integrating textual information into vision tasks. However, in the RSCD domain, limited efforts have been made to effectively leverage textual cues, and challenges persist in capturing differential features. To address these issues, this study proposes the multimodal difference augmentation learning change detection network (MdaCD) that fully exploits textual information and enhances differential feature learning (DFL). MdaCD introduces a CLIP-guided masking process to direct textual descriptions toward image differences, and a multimodal fusion and difference augmentation process to integrate and refine differential features across modalities. The CLIP-guided masking process applies masking to bitemporal image pairs before generating text prompts, enabling a more targeted analysis of changes. Meanwhile, the multimodal fusion and difference augmentation process computes a fused attention map to integrate visual and textual cues, effectively amplifying relevant differences. By applying difference augmentation functions, the differential features from both visual and textual embeddings are further refined and strengthened. The effectiveness of the proposed MdaCD model is validated through extensive experiments on two public RSCD datasets, where it achieves state-of-the-art performance with IoU scores of 84.88% on LEVIR-CD and 71.96% on SYSU-CD. The code and pretrained models of this work will be publicly available at https://github.com/haoyangofficial/MdaCD**

***Index Terms*—Change detection (CD), differential feature learning (DFL), multimodal learning, remote sensing.**

I. Introduction

CHANGE detection (CD) in remote sensing focuses on identifying and labeling areas of change between pairs of remote sensing images [1], which plays a critical role in fields such as environmental monitoring [2], urban planning [3], and disaster recovery [4]. Although the specific objectives of CD may vary across different applications, its overarching goal is to accurately identify and label areas of change within the targeted region of interest [5]. The success of CD depends on several factors, including image quality, temporal resolution (the time between image acquisitions), and the choice of analytical methods [6]. Preprocessing steps, such as radiometric and geometric corrections, play a vital role in ensuring that input images are comparable. Postclassification comparison and direct pixel-based methods are among the approaches commonly used to detect changes [7].

Traditional CD methods for remote sensing image pairs often rely on Siamese neural networks [8], which use a single network with shared weights to extract features from bitemporal image pairs [9]. However, the Siamese network faces significant limitations, particularly when dealing with domain shifts caused by variations in imaging conditions, such as changes in lighting, seasonal effects, atmospheric disturbances, or sensor noise [10]. To address these limitations, other single-modality CD methods have been proposed. Some of these methods employ separate networks to extract features from each image independently, offering greater flexibility to handle domain-specific characteristics [11]. While this strategy can mitigate some issues related to domain shifts, it introduces new challenges, particularly in aligning features effectively between the two images [12]. In a word, although single-modality CD methods, including the Siamese network and its alternatives, have demonstrated some success, their performance is often constrained by inherent limitations [10].

Multimodal methods have shown great potential in overcoming the shortcomings of traditional methods by leveraging complementary information from different modalities [13]. In the field of computer vision, multimodal learning has gained significant traction, with notable advancements in tasks such as image captioning, visual question answering [14], and cross-modal retrieval [15]. For example, models like Visual-BERT, proposed by Devlin et al. [16], extend traditional text-based models by introducing a two-stream architecture to encode both images and text, enabling rich semantic representation through multimodal interactions. Despite the rapid development of multimodal methods in general computer vision tasks, their application in the field of remote sensing is still in its early stages. While models like CLIP [15] have demonstrated remarkable zero-shot adaptability and image recognition capabilities in remote sensing, the use of multimodal methods specifically for CD remains limited. In fact, ChangeCLIP, which is proposed by Dong et al. [17], is currently the only notable attempt to apply multimodal learning in remote sensing change detection (RSCD). ChangeCLIP leverages the powerful textual–visual representation capabilities of the CLIP model to achieve robust performance, effectively bridging the

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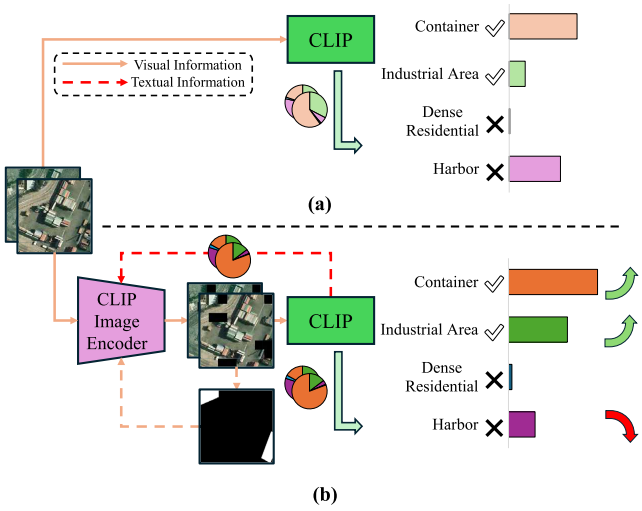


Fig. 1. Textual descriptions generated by the CLIP model, along with the corresponding original and masked bitemporal image pairs, demonstrate that changes in the images can lead to unexplainable impacts in the generated descriptions. The absolute difference between the descriptions is shown in the bar chart, while the text descriptions generated from the original and masked image pairs are represented in the first and second columns, respectively. The changed contexts are marked with a check, while the unchanged contexts are marked with a cross, indicating that the text descriptions generated from the masked image pair emphasize the changed areas more effectively. (a) No CLIP-guided masking. (b) CLIP-guided masking.

gap between textual descriptions and image-based change analysis.

However, multimodal methods still face several shortcomings that limit their performance [18]. In general fields, one significant challenge is the difficulty of accurately aligning information across different modalities, especially when one modality dominates or introduces noise [19]. This issue is particularly pronounced in tasks where textual descriptions fail to capture localized or nuanced changes in the corresponding images [20]. Fig. 1 illustrates these challenges using a bitemporal image pair along with the textual descriptions generated by the CLIP model. The image pair was captured in a harbor, where the only changes are a decrease in sea area and a newly built building in the industrial zone. However, the textual descriptions generated by the CLIP model are unexplainable, as they highlight a decrease in the harbor area while overlooking the change in the industrial area. Two primary factors contribute to this limitation. First, the changed area typically represents only a small portion of the entire image, causing the textual descriptions to be heavily influenced by unchanged regions. Second, domain shifts caused by variations in image acquisition—such as differences in lighting, angles, or sensor properties—further reduce the reliability of the textual descriptions. These issues highlight the need for more robust methods that can better align textual and visual information, particularly in the context of CD.

Beyond these general challenges, two additional open issues remain underexplored in RSCD tasks. First, conventional dual-network architectures for domain adaptation often employ parallel feature encoding but neglect the intrinsic correlations between bitemporal images. Consequently, these models struggle to effectively emphasize change regions, leading to suboptimal change localization and increased sensitivity to background noise. Second, existing difference enhancement techniques tend to focus on modality-specific optimizations, either visual or textual, without achieving synergistic amplification across modalities. Under complex environmental conditions, this lack of cross-modal synergy limits the model's ability to generate geometrically consistent and semantically coherent change maps.

To address these issues, we propose a multimodal difference augmentation learning change detection network (MdaCD), which enhances CD performance by augmenting changes throughout the entire process. MdaCD consists of four key steps: CLIP-guided masking, multimodal encoding, multimodal fusion and difference augmentation, and multimodal decoding. CLIP-guided masking selects blocks with high probabilities of change and extracts their features. Using these features, in the multimodal encoding step, both textual and visual information are encoded into embeddings. In the multimodal fusion and difference augmentation step, the textual and visual embeddings are calculated and integrated across multiple scales and layers. Finally, in the multimodal decoding step, the combined embeddings are decoded to generate the change mask map. The main contributions of this work are summarized as follows.

1) This work proposes a MdaCD that suppresses unchanged regions and enhances differential features across modalities, achieving improved robustness to domain shifts and more effective feature discrimination.
2) A CLIP-guided masking mechanism is designed to refine CLIP outputs by jointly leveraging ground-truth and CLIP supervision. This step suppresses background noise and directs attention toward change regions, thereby alleviating domain shift effects.
3) This work introduces a multimodal fusion and difference augmentation process, which is developed with dual-channel enhancement pipelines guided by fused attention. By augmenting visual–textual discrepancy patterns and cross-modal interactions, it produces geometrically consistent and semantically coherent change maps.

The remainder of the article is structured as follows. Section II reviews several related algorithms and networks. Section III presents the overall architecture of MdaCD, detailing each of its components. In Section IV, we evaluate MdaCD through experiments, comparing it with state-of-the-art algorithms to confirm its effectiveness, followed by an ablation study and a parameter study. Finally, Section V provides a comprehensive summary of MdaCD.

II. Related Works

A. Premasking in RSCD

Premasking is a technique designed to filter out irrelevant or noisy regions of an image prior to performing a CD task [21]. Premasking is particularly beneficial in remote sensing, where vast portions of an image may be dominated

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by unchanged features, such as forests, water bodies, or urban landscapes. By removing these noninformative areas, the computational cost is reduced, and the model's performance, particularly in terms of precision and recall, is improved [22].

Premasking has been applied to enhance feature learning by focusing on informative regions while reducing background interference. One effective implementation is the hard patches mining (HPM) method introduced by Wang et al. [23], which refines this approach by identifying and emphasizing challenging image patches during masked image modeling. By dynamically adjusting mask distributions based on reconstruction difficulty, HPM directs the model's attention to more ambiguous and informative regions, leading to more robust representations. This technique has demonstrated strong potential in feature learning. However, despite their advantages, premasking methods also pose challenges. Improperly defined masks can inadvertently discard useful contextual information, limiting the model's generalization ability. Additionally, designing optimal mask distributions remains a complex task, as overly aggressive masking strategies may hinder rather than enhance feature learning [21].

To address these shortcomings, we introduce CLIP-guided masking to enhance CD by generating a masked image pair for text description generation. Since the unmasked image pair is still used as input to the visual encoder, no valuable visual information is discarded. Furthermore, through ablation and parameter experiments, we determine the optimal masking configuration for each dataset, which further improves the model's ability to learn differential features, enabling the model to outperform existing text description generation methods.

B. DFL in RSCD

Differential feature learning (DFL) focuses on extracting discriminative features that highlight differences between bitemporal image pairs, making it a fundamental technique in RSCD [24]. Unlike methods that directly analyze raw or simple differentiated features, DFL emphasizes learning representations that effectively capture change-specific information while suppressing irrelevant noise, such as background consistency or sensor variations [25]. This approach has proven especially beneficial in detecting subtle changes, such as minor vegetation loss or small-scale urban modifications. By treating differential features as a downstream task, DFL leverages preextracted temporal features and applies advanced strategies like attention mechanisms or feature fusion to enhance its robustness and accuracy [26]. Its ability to focus on change-relevant information makes it a widely adopted paradigm in modern RSCD research [27].

DFL in RSCD faces limitations despite recent advancements. While various methods enhance feature extraction and robustness, challenges remain in effectively capturing temporal differences and mitigating pseudo-change misclassification. For example, Liu et al.'s [27] local-global pyramid network (LGPNet) improves detection performance but may still struggle with generalization. Chang et al.'s [28] Dsfer-Net enhances semantic understanding, yet its reliance on deep supervision can increase computational complexity. Similarly, Guo et al.'s [29] metric learning approach reduces misclassification but requires careful sample selection. Methods like Jia et al.'s [30] SMDNet and Huang et al.'s [31] MFDS-Net introduce attention mechanisms and feature integration modules, yet their effectiveness is constrained by model design and dataset variability. Dong et al.'s [17] ChangeCLIP introduces a differential features compensation module which utilizes various computational methods to help the model learn the differential features effectively, but still faces limitations in performance. These limitations highlight the need for further refinement in DFL-based RSCD.

To address these shortcomings, we propose multimodal fusion and difference augmentation to enhance DFL by explicitly amplifying differences in both visual and textual representations. This process employs a difference augment (DA) function that integrates absolute feature differences, cosine similarity, and multiscale information to construct refined differential representations. By embedding the DA function within a deep learning framework, multimodal fusion and difference augmentation improves the separation of change-relevant features from background noise, resulting in enhanced robustness and accuracy. Additionally, it generates a coarse change mask by computing fused attention through the integration of visual and textual information. Extensive experiments demonstrate that this approach significantly outperforms existing DFL-based methods, offering superior adaptability in complex RSCD scenarios with minimal structural modifications.

III. Proposed Method

A. Multimodal Difference Augmentation Learning Change Detection Network

In this study, we propose a MdaCD, which enhances CD performance by augmenting changes throughout the entire process. As illustrated in Fig. 2, the MdaCD framework consists of four key steps: CLIP-guided masking, multimodal encoding, multimodal fusion and difference augmentation, and multimodal decoding.

In CLIP-guided masking, a Residual Network 50 (ResNet50) [32] is employed as the masking encoder to produce a mask that highlights blocks with a high likelihood of remaining unchanged. Using the masked bitemporal image pairs, CLIP generates corresponding textual descriptions. In multimodal encoding, the original image pairs are processed by the CLIP image encoder to produce multiscale visual embeddings, while the textual descriptions are converted into textual embeddings using the CLIP textual encoder. The multimodal fusion and difference augmentation enhances the differences in both visual and textual embeddings and fuses the information across modalities. Finally, in multimodal decoding, the augmented and fused embeddings are processed by a Swin Transformer decoder [33] equipped with a low-rank bilinear attention mechanism [34] to generate the final change mask, which serves as the network's output.

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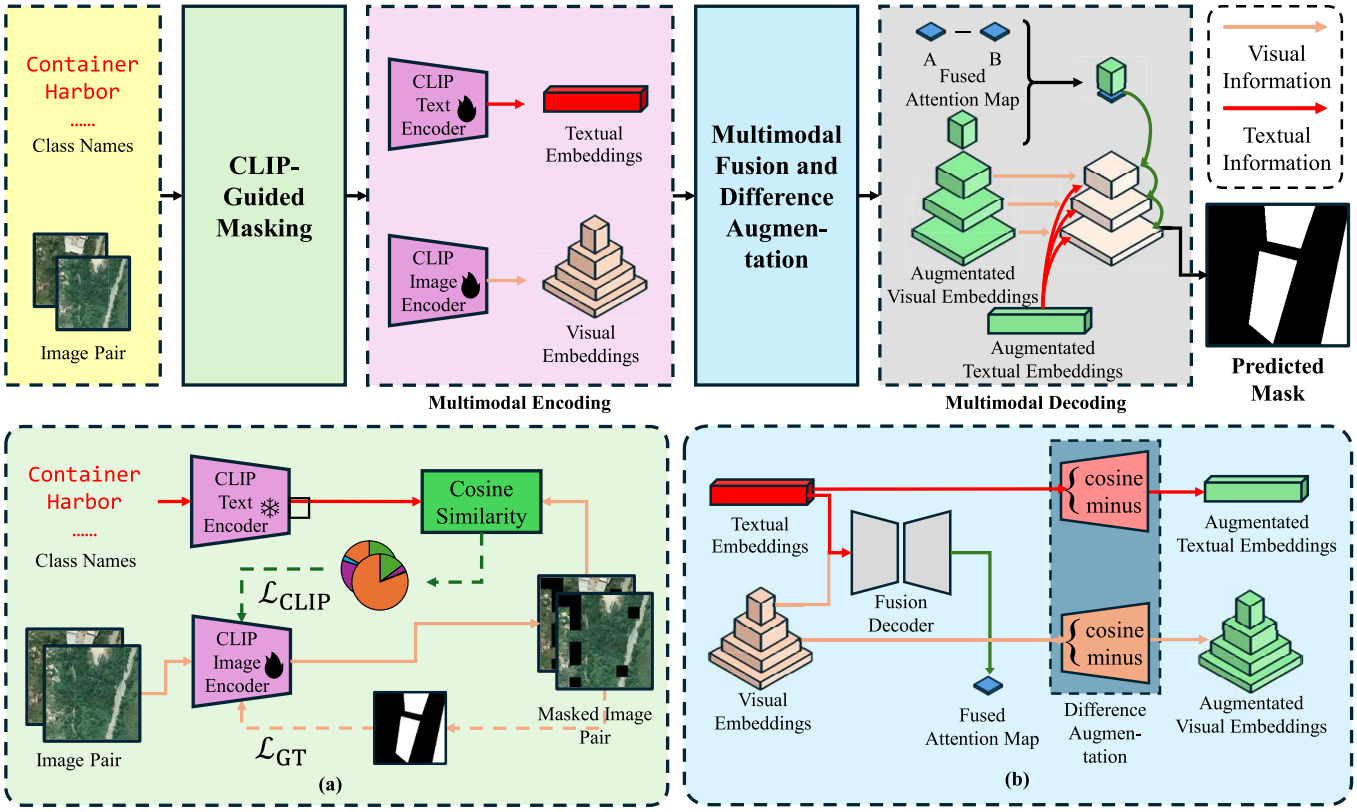


Fig. 2. Overview of the proposed MdaCD. Workflow of (a) CLIP-guided masking process and (b) multimodal fusion and difference augmentation process.

B. CLIP-Guided Masking

As discussed, using the text descriptions generated by CLIP from the original image pair can degrade performance. This is because the unchanged region, which constitutes the majority of the image, strongly influences the output, and variations in imaging conditions can introduce a domain shift. We hypothesize that masking the unchanged region can help CLIP focus on the changed area, thereby improving performance. Therefore, we introduce CLIP-guided masking. This process inputs the bitemporal image pair, generates the mask and text description using the CLIP image encoder (ResNet50 backbone) and a pretrained CLIP model, and outputs the masked image pair and text prompt generated by the text description.

The complete process of CLIP-guided masking is as follows. After computing the difference between the two images, the result is passed through a ResNet up to its penultimate layer, yielding a feature map with a resolution of 1/64 of the original. This feature map is then processed by a 1×1 convolution layer, followed by batch normalization and an activation function, to produce the mask. The generated mask is upsampled to match the resolution of the original image pair, and the masked image pair is obtained by performing element-wise multiplication between the original image pair and the rescaled mask.

Let the original prechange and postchange image be denoted as $\mathbf{x}^1$, $\mathbf{x}^2$, respectively, and the masked image pair as $\mathbf{x}_m^1$ and $\mathbf{x}_m^2$. Let the element-wise multiplication operation be represented by $\odot$. The calculation steps can be expressed as follows:

$$\mathbf{x}_m^k = \mathbf{x}^k \odot \mathbf{M} \tag{1}$$

where

$$\mathbf{M} = \sigma\left(\text{ResNet50}\left(\mathbf{x}^1 - \mathbf{x}^2\right)\right).$$

Here, ResNet50 represents the operation of a ResNet50 up to its penultimate layer, σ represents the operations of the convolution, batch normalization, and activation layers, and $k = 1, 2$ corresponds to the two image acquisition times, respectively.

To mask as much of the unchanged area as possible while preserving the changed regions, we train the image encoder under the supervision of both the ground-truth and CLIP-generated textual information. The loss function ($\mathcal{L}$) is defined as follows:

$$\mathcal{L} = \mathcal{L}_{GT} + w_{CLIP}\mathcal{L}_{CLIP} \tag{2}$$

where $\mathcal{L}_{GT}$ represents the loss based on the ground-truth mask, and $\mathcal{L}_{CLIP}$ represents the loss derived from CLIP textual supervision, and w_{CLIP} is the weight for CLIP loss.

The loss derived from CLIP textual supervision ($\mathcal{L}_{CLIP}$) is to maximize the difference between the CLIP textual information of the masked image pair. Let t_s^k represents the similarity between the image and the textual description s, $\mathbf{S}$ is a default set with 56 different text descriptions, $s \in \mathbf{S}$, and $k = 1, 2$

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correspond to the two image acquisition times, respectively. Then $\mathcal{L}_{\text{CLIP}}$ is defined as follows:

$$\mathcal{L}_{\text{CLIP}} = -\sum_{\forall s \in \mathbf{S}} \left(t_s^1 - t_s^2\right)^2. \tag{3}$$

The loss based on the ground-truth mask ($\mathcal{L}_{\text{GT}}$) aims to minimize errors in masking the changed areas of the image pair. Therefore, $\mathcal{L}_{\text{GT}}$ consists of two components: the loss for the positive region of the ground-truth mask ($\mathcal{L}_p$) and the loss for the negative region of the ground-truth mask ($\mathcal{L}_n$). To ensure the model prioritizes preserving changed regions, $\mathcal{L}_p$ is assigned a greater penalty weight than $\mathcal{L}_n$. Let

$$\mathbf{G} = \{g_{i,j}\}, \mathbf{M} = \{m_{i,j}\} \tag{4}$$

represent the ground-truth change mask and the change mask output by the model, respectively, and the original resolution is $r \times r$, and $i, j \in [1, r]$. The loss based on the ground-truth mask $\mathcal{L}_{\text{GT}}$ and its components are defined as follows:

$$\mathcal{L}_{\text{GT}} = w_p \mathcal{L}_p + \mathcal{L}_n \tag{5}$$

where

$$\begin{cases} \mathcal{L}_p &= \sum_{\forall i,j, g_{i,j}=1} \left(g_{i,j} - m_{i,j}\right)^2 \\ \mathcal{L}_n &= \sum_{\forall i,j, g_{i,j}=0} \left(g_{i,j} - m_{i,j}\right)^2. \end{cases}$$

In this case, $\mathcal{L}_p$ represents the loss for the positive region of the ground-truth mask, and $\mathcal{L}_n$ represents the loss for the negative region of the ground-truth mask. As the primary objective is to retain all blocks with potential changes, the penalty weight for the positive loss, w_p, is set much higher than 1.

Next, the masked image pair is used to generate the text prompt. Using the masked image pair, the CLIP model produces a pair of textual descriptions that represent the nature of each image. From these descriptions, we identify the nine most prominent classes that represent the majority of the content. These class names are then incorporated into structured sentences, forming the text prompt. Let these class names be $c_1, c_2, \ldots, c_9$. Then the structured sentence can be formulated as follows:

$$\{\text{Foreground}\} + c_1 + c_2 +, \cdots, +c_9 + \{\text{Background}\} \tag{6}$$

where {Foreground} and {Background} are fixed textual templates used to integrate the class names into a coherent sentence. The original image pair $\mathbf{x}^1, \mathbf{x}^2$ and the sentence serve as inputs to multimodal encoding, producing corresponding visual and textual embeddings. The main steps of CLIP-guided masking are shown in Fig. 2(a).

C. Multimodal Fusion and Difference Augmentation

First, to enhance the interaction between visual and textual embeddings, a fused attention map is generated by integrating both modalities, helping to retain the original information before applying augmentations. Then, to better capture the differential features, the encoder-extracted features are further enhanced. To achieve this, we define a DA function that amplifies the differences in both visual and textual information processes. Let the features of the prechange and postchange image $(\mathbf{x}^1, \mathbf{x}^2)$ extracted from the CLIP image encoder be denoted as $\mathbf{F}^1$, $\mathbf{F}^2$. Initially, these features are concatenated. Subsequently, additional operations are performed, including the absolute difference ($\mathbf{F}_{\text{minus}}$) and the cosine similarity ($m_{\mathbf{F}}$), both of which are designed to capture complementary aspects of change. The results of these computations are concatenated with the original features to enrich the representation, and channel attention is applied to emphasize informative channels. This process enables more adequate extraction of differential features across modalities. The complete workflow for computing $\text{DA}(\mathbf{F}^1, \mathbf{F}^2)$ is as follows:

$$\text{DA}\left(\mathbf{F}^1, \mathbf{F}^2\right) = \text{CA}\left([\mathbf{F}, \mathbf{F}_{\text{minus}}, \mathbf{F} \cdot m_{\mathbf{F}}]\right) \tag{7}$$

where

$$\begin{cases} \mathbf{F} = \left[\mathbf{F}^1, \mathbf{F}^2\right] \\ \mathbf{F}_{\text{minus}} = \text{Conv}\left(|\mathbf{F}^1 - \mathbf{F}^2|\right) \\ m_{\mathbf{F}} = \sigma\left(\mathbf{F}^1, \mathbf{F}^2\right). \end{cases}$$

Here, Conv represents a 1×1 convolution layer, CA represents the channel attention, $[\cdot]$ denotes the concatenation, σ represents the cosine similarity, activation function, and average value, and $k = 1, 2$ correspond to the two image acquisition times, respectively.

Then, we use a multiscale visual integrator to augment and combine the multiscale visual features extracted in multimodal encoding. The first three layers of the feature pyramid, each downscaled to 1/64 of the original resolution, are concatenated and subsequently passed through a 1×1 convolution layer to reduce the channel dimensionality. The last layer is processed through a DA function. Finally, the results from these two operations are concatenated and passed through another 1×1 convolution layer to generate the visual embeddings.

Let $\mathbf{F}^k = \{\mathbf{F}_1^k, \mathbf{F}_2^k, \ldots, \mathbf{F}_n^k\}, k = 1, 2$ represent the output of an n-layer feature pyramid of each image of the image pair generated in multimodal encoding, separately. Here, $k = 1, 2$ correspond to the two image acquisition times, respectively. The process is defined as follows:

$$\mathbf{V} = \text{Conv}\left(\left[\mathbf{F}'_{\text{other}}, \mathbf{F}'_n\right]\right) \tag{8}$$

where

$$\begin{cases} \mathbf{F}'_l = \text{Conv}\left(\text{Rescale}\left(\left[\mathbf{F}_l^1, \mathbf{F}_l^2\right]\right)\right) \\ \mathbf{F}'_{\text{other}} = \text{Conv}\left(\left[\mathbf{F}'_1, \mathbf{F}'_2, \ldots, \mathbf{F}'_{n-1}\right]\right) \\ \mathbf{F}'_n = \text{DA}\left(\left[\mathbf{F}_n^1, \mathbf{F}_n^2\right]\right)). \end{cases}$$

Here, Rescale represents the downscaling operation to 1/64 of the original resolution, DA is the DA function, and $[\cdot]$ denotes the concatenation operation. Each Conv represents a different 1×1 convolution layer, $\mathbf{V}$ represents the final visual embeddings, and $l = 1, 2, \ldots, n-1$. The augmented textual embeddings are obtained by applying the DA function to the textual embeddings produced.

In summary, the visual and textual embeddings first contribute to generating a fused attention map, which highlights potential changed areas based on textual information. These embeddings are then processed through the DA function. The main steps of multimodal fusion and difference augmentation are illustrated in Fig. 2(b).

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TABLE I
QUANTITATIVE RESULTS OF THE EXPERIMENT ON BOTH LEVIR-CD AND SYSU-CD DATASETS. THE BEST RESULT IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS UNDERLINED

Model	Parameters (M)	FLOPs (G)	LEVIR-CD (%)					SYSU-CD (%)				
			IoU	Acc.	Prec.	Rec.	F1	IoU	Acc.	Prec.	Rec.	F1
CD-Lamba [41]	28.74	15.26	79.10	98.83	89.72	86.99	88.33	70.39	91.76	82.15	**83.10**	82.62
CDMask [42]	24.49	32.46	79.05	98.78	86.60	90.07	88.30	69.77	92.02	**86.72**	78.12	82.19
CDXLSTM [43]	16.19	3.92	83.30	99.07	91.52	90.27	90.89	68.70	91.72	86.33	77.09	81.45
ChangeCLIP [17]	-	-	84.71	99.17	92.83	**90.64**	91.72	70.53	92.08	85.16	80.41	82.72
DDLNet [44]	12.67	7.35	78.66	98.79	88.93	87.19	88.05	68.68	91.69	86.07	77.27	81.43
LCDNet [45]	2.56	4.45	51.05	97.15	80.24	58.40	67.60	69.23	91.82	86.04	77.98	81.82
MSCANet [46]	-	-	80.22	98.90	90.65	87.46	91.02	67.72	91.13	82.65	78.95	80.76
SEIFNet [47]	27.91	8.37	83.40	99.09	92.49	89.46	90.95	69.96	91.90	84.81	79.98	82.32
MdaCD (Ours)	141.26	76.23	**84.88**	**99.18**	**93.62**	90.09	**91.82**	**71.96**	**92.45**	85.25	82.20	**83.70**

IV. EXPERIMENTS AND RESULTS

A. Experiment Setup

This study implements the code using PyTorch [35], building on the MMSegmentation library [36]. All experiments are carried out on a system equipped with an Intel Core i5-14600KF processor, 128 GiB of RAM, and two NVIDIA GeForce RTX 3090 GPUs. The operating system is Ubuntu 20.04 (Focal), with CUDA version 12.8. The MdaCD framework employs cross-entropy loss as its loss function [37] and uses the AdamW optimizer [38], with an initial learning rate of 3×10^{-5} and a weight decay of 0.01. Benchmark tests are conducted on two public RSCD datasets: LEVIR-CD [39] and SYSU-CD [40]. Performance is evaluated using five metrics, including interaction over union (IoU), overall accuracy (Acc.), precision (Prec.), recall (Rec.), and $F1$ score ($F1$). The batch size is set to 20 in both datasets. The proposed model is trained for 30 000 iterations on the LEVIR-CD dataset and for 10 000 iterations on the SYSU-CD dataset. Compared to previous approaches, MdaCD achieves strong results in RSCD tasks.

B. Experimental Datasets Description

To demonstrate the superiority of MdaCD, the experiments were conducted on two RSCD datasets: LEVIR-CD and SYSU-CD. In this study, the LEVIR-CD dataset is preprocessed before conducting experiments. The following is the detailed information on the datasets:

The LEVIR-CD dataset consists of 637 pairs of images with a very high resolution of 0.5 m and a size of 1024×1024, captured by Chen et al. [39] using Google Earth. These bitemporal images with a time span of 5 to 14 years have significant land-use changes, especially the construction growth. The dataset is divided into a training set of 445 image pairs, a validation set of 64 image pairs, and a testing set of 128 image pairs. Each image is cropped into 64 slices, each with a size of 256×256, with no overlapping.

The SYSU-CD dataset consists of 20 000 pairs of aerial images with a resolution of 0.5 m and a size of 256×256, captured by Shi et al. [40] in Hong Kong between 2007 and 2014. The primary types of changes in this dataset include new urban construction, suburban expansion, preconstruction foundations, and vegetation changes. In this study, the dataset is divided into 12 000 image pairs for training, 4000 image pairs for validation, and 4000 image pairs for testing.

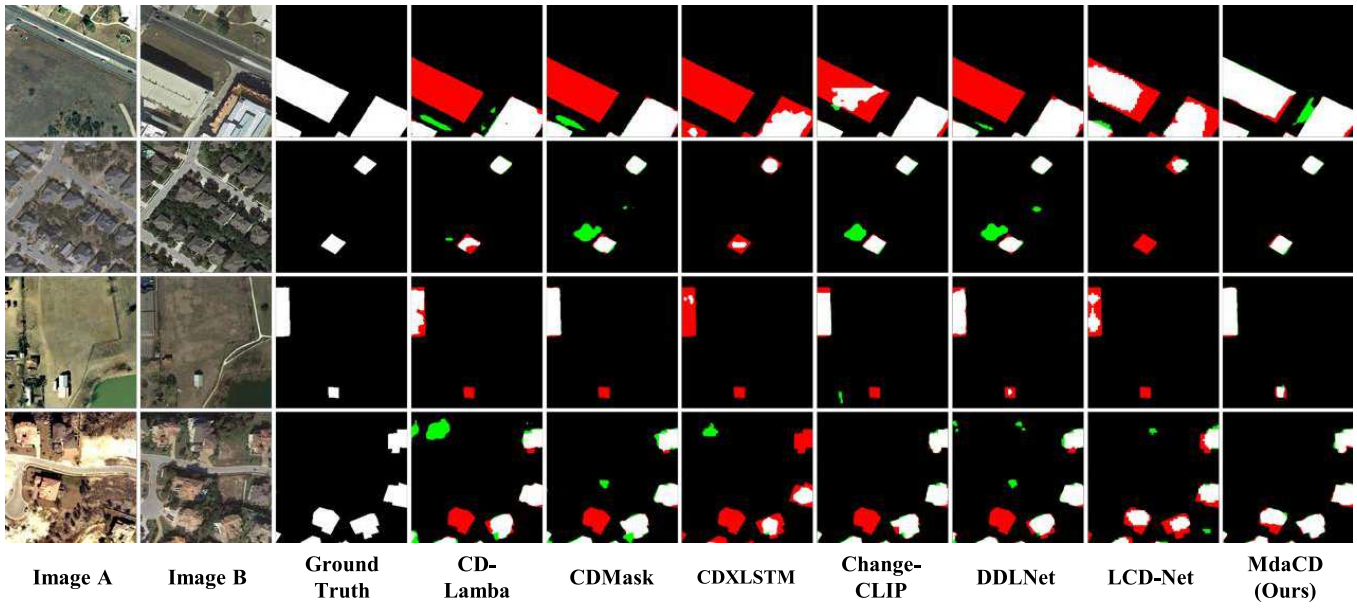
C. Comparisons With Previous Methods

We compared the performance of MdaCD with other state-of-the-art (SOTA) methods from 2024 and 2025, including Wu et al.'s [41] CD-Lamba, Ma et al.'s [42] CDMask, Wu et al.'s [43] CDXLSTM, Dong et al.'s [17] ChangeCLIP, Ma et al.'s [44] DDLNet, Liu et al.'s [45] LCD-Net, Zheng et al.'s [46] MSCANet, and Huang et al.'s [47] SEIFNet.

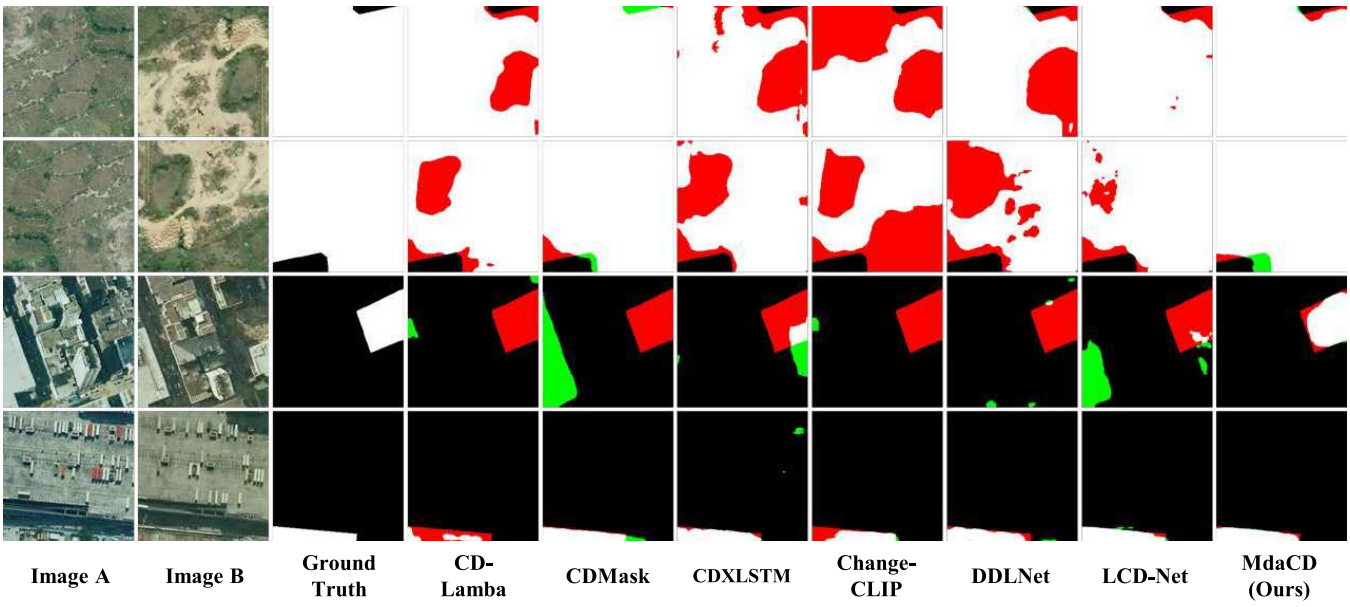
MdaCD achieved state-of-the-art performance across both datasets according to various evaluation metrics, as shown in Table I. We retrained and evaluated several methods using the introduced preprocessing pipeline, as the original papers employed varying preprocessing strategies. In the LEVIR-CD dataset, MdaCD achieved an IoU index of 84.88%, with the overall accuracy (99.18%), precision (93.62%), and $F1$ score (91.82%) all reaching state-of-the-art levels compared to other methods. In the SYSU-CD dataset, MdaCD achieved an IoU index of 71.96%, with the overall accuracy (92.45%) and $F1$ score (83.70%) all reaching state-of-the-art levels compared to other methods. From the tables, we conclude that MdaCD achieves the highest IoU index across both datasets while maintaining a superior $F1$ score, consistently outperforming the second-best method.

The visualization of comparisons with other SOTA methods on the LEVIR-CD and SYSU-CD dataset is shown in Fig. 3, where true positive (TP), true negative (TN), false positive (FP), and false negative (FN) are represented in white, black, green, and red, respectively. From the first three rows in Fig. 3(a), we observe that MdaCD successfully detects all changed objects, whereas other state-of-the-art methods either miss some changes or exhibit poor segmentation performance. In the last row, MdaCD shows moderate success in identifying the house change in the bottom left corner, which is likely affected by domain variation in the image pair. Similarly, the first two rows in Fig. 3(b) illustrate that MdaCD effectively detects change regions in textual contexts. Its strong performance in urban scenes is evident in the last two rows. Overall, the figure highlights that, compared to several leading methods from the past two years, MdaCD achieves superior precision and robust overall performance.

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(a)



(b)

Fig. 3. Qualitative results of different methods on both (a) LEVIR-CD and (b) SYSU-CD dataset. TP, TN, FP, and FN are represented in white, black, green and red, respectively.

TABLE II

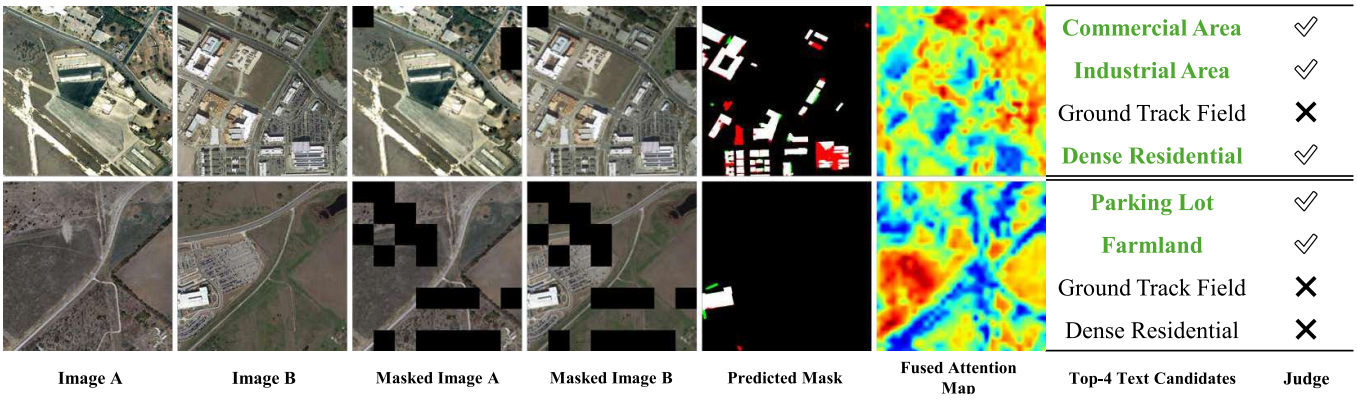
QUANTITATIVE RESULTS OF THE ABLATION EXPERIMENT ON BOTH LEVIR-CD AND SYSU-CD DATASETS. THE BEST RESULT IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS UNDERLINED. UNIT: PERCENTAGE (%)

Model	LEVIR-CD					SYSU-CD				
	IoU	Acc.	Prec.	Rec.	F1	IoU	Acc.	Prec.	Rec.	F1
Baseline	84.17	99.14	92.68	90.17	91.41	69.93	91.79	83.68	80.98	82.31
Baseline + CLIP-Guided Masking	84.86	99.17	92.82	**90.82**	91.81	71.79	**92.52**	**86.68**	80.69	83.58
Baseline + Multimodal Fusion and Difference Augmentation	84.58	99.16	93.46	89.90	91.64	70.94	92.25	85.98	80.22	83.00
Full	**84.88**	**99.18**	**93.62**	90.09	**91.82**	**71.96**	92.45	85.25	**82.20**	**83.70**

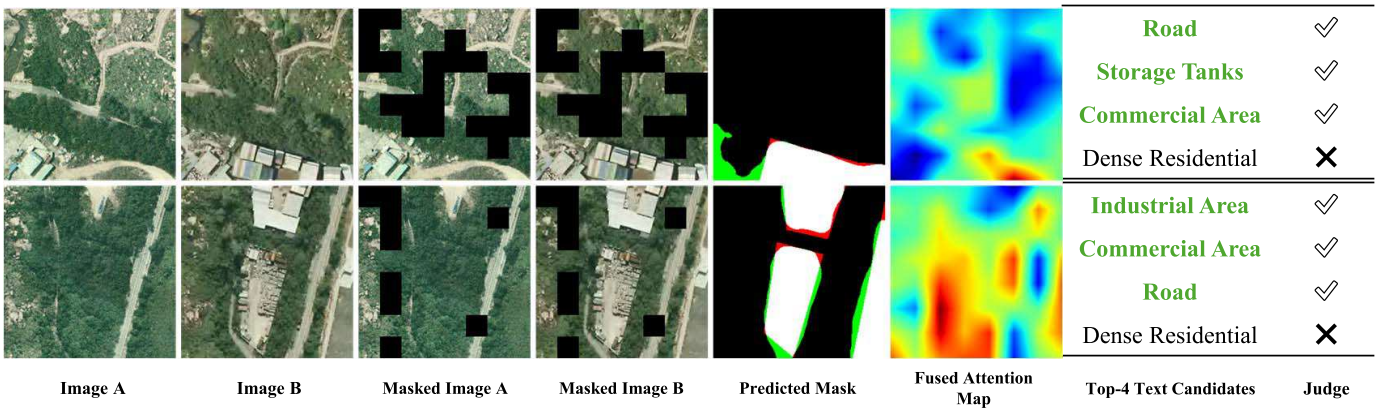
D. Ablation Study

In this study, ablation experiments are conducted on both datasets with results presented in Table II. We first construct a baseline model by removing CLIP-guided masking and multimodal fusion and difference augmentation. The baseline model uses the text description generated from the unmasked

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(a)



(b)

Fig. 4. Qualitative results of the masked image pairs, fused attention maps, and the top four text candidates on both (a) LEVIR-CD and (b) SYSU-CD datasets. For the predicted masks, TP, TN, FP, and FN are represented in white, black, green, and red, respectively. The top four text candidates with the greatest changes in textual information of the image pair are highlighted in bold and green, with a check to see if the text represents a changed context. Otherwise, they are marked with a cross if the text refers to an unchanged context or does not exist.

image pairs, uses concatenation instead of DA functions in multimodal fusion and difference augmentation, and drops out the fused attention map difference in multimodal fusion and difference augmentation. After evaluating the baseline model, we incrementally add CLIP-guided masking and Multimodal Fusion, Difference Augmentation, and test the performance separately.

The results in Table II demonstrate that each step contributes to improving the metrics, confirming their effectiveness in enhancing the overall performance of the MdaCD model. Notably, CLIP-guided masking has a greater impact on performance improvement compared to multimodal fusion and difference augmentation. With CLIP-guided masking, MdaCD achieved an IoU of 84.86% on the LEVIR-CD dataset and 71.79% on the SYSU-CD dataset. Notably, it attained a higher recall than the complete MdaCD model on LEVIR-CD and a higher precision than the full model on SYSU-CD. Incorporating multimodal fusion and difference augmentation further enhanced performance across all metrics except recall on LEVIR-CD and precision on SYSU-CD. While a trade-off between precision and recall is observed, the overall improvement is evident through the increase in $F1$ score.

The visualization results of the masked image pairs, fused attention maps, and the top four text candidates showing the greatest changes in textual information for both the LEVIR-CD and SYSU-CD datasets are presented in Fig. 4. From the masked image pairs, it is evident that CLIP-guided masking effectively excludes the changed regions while suppressing unchanged contexts. Additional examples are provided in Fig. 5, demonstrating that CLIP-guided masking consistently increases the confidence scores of changed contexts while reducing those of unchanged or nonexistent contexts. In realistic scenarios, objects frequently appear in both changed and unchanged regions. As shown in Fig. 6, CLIP-guided masking continues to deliver strong improvements under such complex conditions. Word clouds of the text descriptions generated by the CLIP model, with and without CLIP-guided masking, are displayed in Fig. 7. The masked cases exhibit greater lexical diversity than the unmasked ones, suggesting that CLIP-guided masking effectively uncovers and amplifies a wider range of contextual information. The fused attention maps reveal that MdaCD forms an initial understanding of potential change areas; however, it must still determine whether a detected change is sufficiently relevant to be represented in the final output mask (e.g., changes involving a parking lot). Analysis of the text candidates indicates that the changed context is almost always identified, but the model must further decide whether it should be regarded as a meaningful change.

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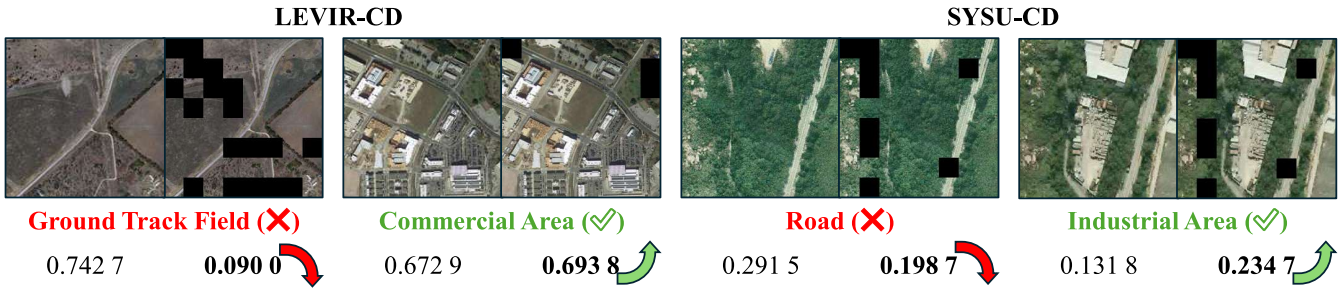


Fig. 5. Text context with the highest confidence level and its corresponding confidence score (displayed below the images) for both the original and masked images from the LEVIR-CD and SYSU-CD datasets. The text context is marked in green with a check if it represents the changed context and in red with a cross if it does not. The confidence level of the masked images is highlighted in bold, indicating a stronger alignment with the actual changes.

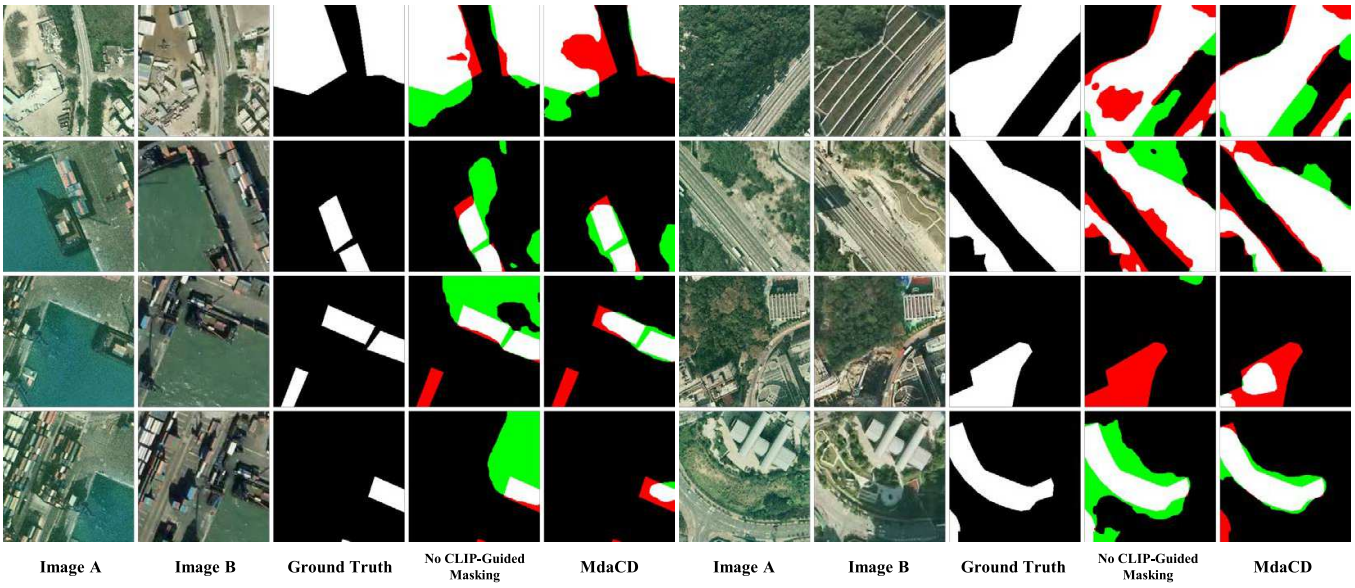


Fig. 6. Qualitative ablation results of MdaCD on the SYSU-CD dataset. In the predicted masks, TPs are shown in white, TNs in black, FPs in green, and FNs in red. With CLIP-guided masking, MdaCD demonstrates improved performance in more realistic scenarios, where objects appear in both changed and unchanged regions.

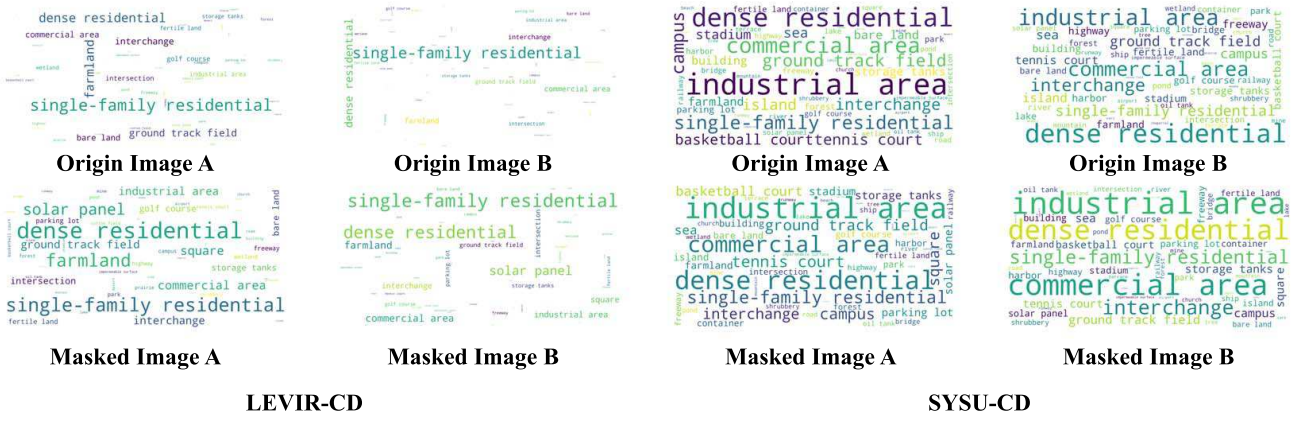


Fig. 7. Word clouds of the text descriptions generated by CLIP from the original and masked image pairs in both the LEVIR-CD and SYSU-CD datasets. Compared to the original descriptions, those from the masked image pairs exhibit greater diversity and a wider range of contexts.

E. Parameter Study

Next, we conducted parameter experiments on CLIP-guided masking, testing different penalty weight values for the positive ground-truth loss (w_p) and the CLIP loss (w_{CLIP}). Specifically, we experimented with two penalty weight values for the positive loss: 10 and 100, without the loss of the CLIP-generated textual information (as $w_{\text{CLIP}} = 0$). After

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TABLE III

QUANTITATIVE RESULTS OF THE PARAMETER EXPERIMENT ON BOTH LEVIR-CD DATASET AND SYSU-CD DATASET. THE BEST RESULT IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS UNDERLINED

Loss Weights		LEVIR-CD (%)					SYSU-CD (%)				
w_p	w_{CLIP}	IoU	Acc.	Prec.	Rec.	F1	IoU	Acc.	Prec.	Rec.	F1
10	0	84.79	**99.18**	93.39	90.21	91.77	71.66	92.33	84.74	**82.27**	83.49
100	0	84.45	99.16	93.54	89.68	91.57	71.18	92.38	**86.83**	79.80	83.17
10	10^2	**84.86**	99.17	92.82	**90.82**	**91.81**	71.45	92.41	86.36	80.53	83.35
10	10^3	84.59	99.17	**93.69**	89.71	91.65	**71.79**	**92.52**	86.68	80.69	**83.58**

finding the best-performed w_p value, we tested two w_{CLIP} values: 10^2 and 10^3, based on that w_p value. Experiments were conducted across both datasets, and the results are presented in Table III.

The results presented in Table III demonstrate that setting the penalty weight $w_p = 10$ yields the most effective performance across both the LEVIR-CD and SYSU-CD datasets. For the LEVIR-CD dataset, the best overall performance is achieved when $w_{CLIP} = 10^2$, with the highest IoU (84.86%), $F1$ score (91.81%), and Recall (90.82%). This configuration also maintains competitive Precision (92.82%) and Accuracy (99.17%). On the SYSU-CD dataset, the best configuration is obtained with $w_{CLIP} = 10^3$, leading to the highest IoU (71.79%) and $F1$ score (83.58%), as well as strong results across all other metrics. While the baseline configuration ($w_{CLIP} = 0$) already performs well, particularly for $w_p = 10$, the addition of CLIP guidance further refines the results, especially when carefully tuned.

These trends suggest that the impact of CLIP guidance varies depending on dataset characteristics. The SYSU-CD dataset, which includes a broader range of change types and more complex scene variations, benefits from a stronger influence of CLIP (i.e., higher w_{CLIP}), likely due to CLIP's ability to incorporate richer semantic understanding during the CD process. In contrast, the more structurally consistent LEVIR-CD dataset reaches optimal performance with moderate CLIP weight. Moreover, the relatively small performance differences between configurations with different w_p values indicate that while penalty weighting influences performance, its effect is secondary to the presence and tuning of the CLIP-guided masking mechanism. This highlights the importance of leveraging semantic-level cues provided by CLIP, especially in diverse or complex data environments.

V. CONCLUSION

In this work, we propose MdaCD, a model designed to detect changes in remote sensing images by enhancing differences in both visual and textual information. We introduce a CLIP-guided masking process, which utilizes part of the CLIP image encoder network to generate a mask prior to CD. The masked image pairs are then processed by the CLIP model to generate text prompts. We demonstrate its strong capability in amplifying changed contexts while suppressing unchanged ones through various visualization results. To extract differential features from both visual and textual embeddings, we implement a DA function that enhances these features and amplifies modality-specific differences. Additionally, we improve the fusion of multimodal embeddings by computing a fused attention map, providing the model with a coarse understanding of the changed area.

To evaluate the performance of MdaCD, we conduct experiments on two public remote sensing CD datasets: LEVIR-CD and SYSU-CD. Quantitative results show that MdaCD achieves state-of-the-art performance in terms of IoU across both datasets while maintaining a high $F1$ score. An ablation study is also conducted to assess the individual contributions of CLIP-guided masking and Multimodal Fusion, and Difference Augmentation. The results indicate that both components improve overall performance, with CLIP-guided masking having a more substantial impact. Furthermore, a parameter study is carried out to identify optimal loss weights for the CLIP-guided masking process. Based on the results, we conclude that MdaCD achieves state-of-the-art performance in multimodal remote sensing CD tasks.

REFERENCES

[1] G. Cheng et al., "Change detection methods for remote sensing in the last decade: A comprehensive review," *Remote Sens.*, vol. 16, no. 13, p. 2355, Jun. 2024.

[2] C. Liu, J. Zhang, K. Chen, M. Wang, Z. Zou, and Z. Shi, "Remote sensing SpatioTemporal vision-language models: A comprehensive survey," 2024, *arXiv:2412.02573*.

[3] H. Wang and M. Wu, "Urban change detection of remote sensing images via deep-feature extraction," *Sci. Rep.*, vol. 15, no. 1, p. 21473, 2025.

[4] C. Liu, K. Chen, H. Zhang, Z. Qi, Z. Zou, and Z. Shi, "Change-agent: Toward interactive comprehensive remote sensing change interpretation and analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635616.

[5] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.

[6] M. Pourreza, F. Moradi, M. Khosravi, A. Deljouei, and M. K. Vanderhoof, "GCPs-free photogrammetry for estimating tree height and crown diameter in Arizona cypress plantation using UAV-mounted GNSS RTK," *Forests*, vol. 13, no. 11, p. 1905, Nov. 2022.

[7] Q. Demarquet, S. Rapinel, O. Gore, S. Dufour, and L. Hubert-Moy, "Continuous change detection outperforms traditional post-classification change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 133, pp. 104142–104155, 2024.

[8] X. Ou, L. Liu, S. Tan, G. Zhang, W. Li, and B. Tu, "A hyperspectral image change detection framework with self-supervised contrastive learning pretrained model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7724–7740, 2022.

[9] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[10] D. Peng, X. Liu, Y. Zhang, H. Guan, Y. Li, and L. Bruzzone, "Deep learning change detection techniques for optical remote sensing imagery: Status, perspectives and challenges," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104282.

[11] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

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[12] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual task constrained deep Siamese convolutional network model," 2019, *arXiv:1909.07726*.
[13] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.
[14] Z. Gan, L. Li, C. Li, L. Wang, Z. Liu, and J. Gao, "Vision-language pre-training: Basics, recent advances, and future trends," 2022, *arXiv:2210.09263*.
[15] A. Radford et al., "Learning transferable visual models from natural language supervision," 2021, *arXiv:2103.00020*.
[16] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
[17] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.
[18] X. Peng, Y. Wei, A. Deng, D. Wang, and D. Hu, "Balanced multimodal learning via on-the-fly gradient modulation," 2022, *arXiv:2203.15332*.
[19] K. Yang, W.-Y. Lin, M. Barman, F. Condessa, and Z. Kolter, "Defending multimodal fusion models against single-source adversaries," 2022, *arXiv:2206.12714*.
[20] S. Li and H. Tang, "Multimodal alignment and fusion: A survey," 2024, *arXiv:2411.17040*.
[21] F. Xiong, T. Li, J. Chen, J. Zhou, and Y. Qian, "Mask guided local–global attentive network for change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1–14, 2024.
[22] Y. Zhang, L. Zhang, and D. Tao, "Deep learning in remote sensing: A comprehensive review and list of research directions," *IEEE Access*, vol. 7, pp. 62420–62444, 2019.
[23] H. Wang, K. Song, J. Fan, Y. Wang, J. Xie, and Z. Zhang, "Hard patches mining for masked image modeling," 2023, *arXiv:2304.05919*.
[24] Q. Zang, D. Zhao, S. Wang, D. Quan, and Z. Zhong, "Feature spectrum learning for remote sensing change detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 12647–12657.
[25] H. Hou, Y. Wang, Q. Qin, Y. Tan, and T. Liu, "Multi-scale feature fusion based on difference enhancement for remote sensing image change detection," *Symmetry*, vol. 17, no. 4, p. 590, Apr. 2025.
[26] L. Li et al., "Multidirectional attention fusion network for SAR change detection," *Remote Sens.*, vol. 16, no. 19, p. 3590, Sep. 2024.
[27] T. Liu et al., "Building change detection for VHR remote sensing images via local–global pyramid network and cross-task transfer learning strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4704817.
[28] S. Chang, M. Kopp, P. Ghamisi, and B. Du, "Dsfer-Net: A deep supervision and feature retrieval network for bitemporal change detection using modern Hopfield networks," 2023, *arXiv:2304.01101*.
[29] H. Guo, B. Du, C. Wu, C. Han, and L. Zhang, "DeepCL: Deep change feature learning on remote sensing images in the metric space," 2023, *arXiv:2307.12208*.
[30] J. Jia, G. Lee, Z. Wang, L. Zhi, and Y. He, "Siamese meets diffusion network: SMDNet for enhanced change detection in high-resolution RS imagery," 2024, *arXiv:2401.09325*.
[31] Z. Huang, Z. Fu, S. Jintao, G. Yuan, and J. Li, "MFDS-Net: Multi-scale feature depth-supervised network for remote sensing change detection with global semantic and detail information," 2024, *arXiv:2405.01065*.
[32] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[33] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[34] J.-H. Kim, J.-H. Jun, and B. Zhang, "Bilinear attention networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 9992–10002.
[35] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2022, pp. 8026–8037.
[36] *MMSegmentation: OpenMMLab Semantic Segmentation Toolbox and Benchmark*, MMS Contributors, OpenMMLab, China, 2020.
[37] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016, ch. 6.
[38] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2022, pp. 601–613.
[39] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[40] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
[41] Z. Wu et al., "CD-Lamba: Boosting remote sensing change detection via a cross-temporal locally adaptive state space model," 2025, *arXiv:2501.15455*.
[42] X. Ma, Z. Wu, R. Lian, W. Zhang, and S. Song, "Rethinking remote sensing change detection with a mask view," 2024, *arXiv:2406.15320*.
[43] Z. Wu, X. Ma, R. Lian, K. Zheng, and W. Zhang, "CDXLSTM: Boosting remote sensing change detection with extended long short-term memory," 2024, *arXiv:2411.07863*.
[44] X. Ma, J. Yang, R. Che, H. Zhang, and W. Zhang, "DDLNet: Boosting remote sensing change detection with dual-domain learning," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2024, pp. 1–6.
[45] W. Liu, J. Li, H. Wang, R. Tan, Y. Fu, and Q. Tian, "LCD-Net: A lightweight remote sensing change detection network combining feature fusion and gating mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 7769–7783, 2025.
[46] X. Zheng, X. Lin, L. Qing, and X. Ou, "Semantic-aware remote sensing change detection with multi-scale cross-attention," *Sensors*, vol. 25, no. 9, p. 2813, Apr. 2025.
[47] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.

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